SMART INDIA HACKATHON 2026

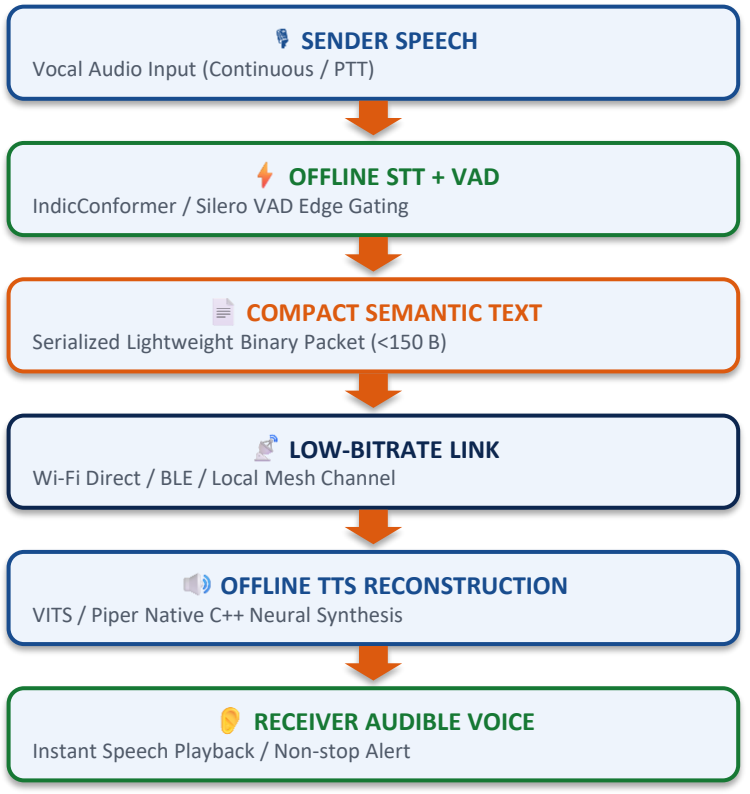
OFFICIAL IDEA PRESENTATION FORMAT | PRELIMINARY ROUND

ISRO PS #26173
SOFTWARE CATEGORY

PROBLEM STATEMENT SUBMISSION DETAILS

- **Problem Statement ID:** 26173
- **Problem Statement Title:** iTantra - Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for Low Bitrate Links
- **Organization:** Indian Space Research Organisation (ISRO)
- **Department:** Department of Space / ISRO
- **Theme:** Smart Automation
- **Category:** Software
- **Team ID:** [TEAM ID]
- **Team Name:** [TEAM NAME]

ACOUSTIC → SEMANTIC → ACOUSTIC



TRANSMIT MEANING. RECONSTRUCT VOICE.

IDEA / PROPOSED SOLUTION

PROBLEM EXISTING

➢ **Data-Intensive Raw Voice:** Uncompressed / standard audio requires high continuous bandwidth (>64 kbps), making transmission impossible over low-bitrate, embedded radio, or congested distress channels.

➢ **Cloud Speech Failure in Disasters:** Existing voice translation and communication tools depend on internet-hosted APIs (Google/AWS), completely failing when local telecom infrastructure collapses.

➢ **Text Inaccessibility in Distress:** Text-only alerts fail non-literate populations who require immediate, actionable voice broadcasts in their regional mother tongues.

PROPOSED SOLUTION: ITANTRA NEURAL TRANSCEIVER

➢ **Acoustic-Semantic-Acoustic Paradigm:** Converts voice to text on transmitter via offline STT, transmits compact binary packets over Wi-Fi/BLE, and reconstructs speech via offline TTS at receiver.

➢ **10 Indian Languages On-Device:** Lightweight, INT8 quantized neural models for Hindi, Marathi, Gujarati, Tamil, Telugu, Kannada, Malayalam, Bengali, Odia & English running 100% locally.

➢ **Dual Operating Modes & Distress Alerts:** Supports Push-to-Talk (Walkie-Talkie), full-duplex phone mode with Acoustic Echo Cancellation (AEC), and non-interruptible maximum-volume emergency broadcasts.

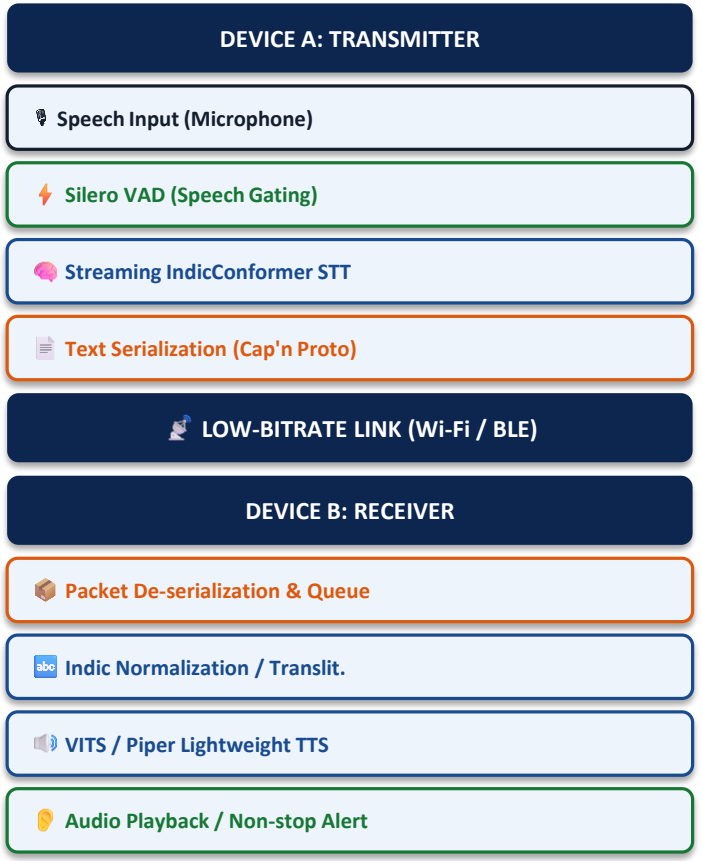
UVP (UNIQUE VALUE PROPOSITION)

➢ **1. Voice Accessibility Without Voice Bandwidth:** Reduces transmission payload to compact text (<150 bytes per sentence), overcoming physical link bitrate bottlenecks.

➢ **2. Zero Cloud Dependency & Edge-First:** Native C++ Sherpa-ONNX runtime and Android AAudio engine operate fully off-grid with zero cloud subscriptions or API latency.

➢ **3. Symmetrical Transceiver Architecture:** Every node functions as both neural sender and receiver with seamless automatic role switching.

TRANSCEIVER ARCHITECTURE



TECHNICAL APPROACH & ARCHITECTURE

END-TO-END ACOUSTIC-SEMANTIC-ACOUSTIC PIPELINE

1. Audio Capture
AAudio Native NDK
16kHz PCM Buffer
Low-latency callback

2. Voice Gating
Silero VAD Gating
Idle STT Sleep
Pause Detection

3. Edge ASR
IndicConformer
Streaming RNN-T/CTC
INT8 Quantized

4. Binary Packet
Cap'n Proto Schema
Sender ID + Lang ID
<150 B / sentence

5. P2P Transport
Wi-Fi Direct / BLE
Local Mesh Bridge
CRC Integrity Check

6. Text Synthesis
VITS / Piper TTS
IndicXlit Phonemes
Sherpa-ONNX Core

7. Voice Playback
AAudio Engine
AEC Loop Filter
Distress Audio Focus

SPEECH-TO-TEXT & VAD GATING

- **IndicConformer ASR:** Selected research direction for streaming Indic ASR; hybrid RNN-T / CTC decoding for instant token emission.
- **Silero VAD CPU Gating:** Lightweight VAD continuously monitors audio; heavy STT model stays dormant during silence (Idle CPU target <2%).
- **Sentence Boundary Logic:** Detects acoustic pauses (~400-600ms) to finalize sentence tokens for immediate low-latency packetization.
- **INT8 & mmap Optimization:** Dynamic quantization & memory mapping reduce memory footprint to ~180MB (research reference).

TEXT-TO-SPEECH & NATIVE AUDIO

- **VITS / Piper Architecture:** Lightweight, end-to-end neural acoustic model synthesized in native C++ via Sherpa-ONNX (RTF target <0.25).
- **Indic Text Normalization:** Rule-based numeral/date expansion and IndicXlit transliteration for high phonetic naturalness across 10 scripts.
- **Android AAudio Engine:** Native NDK C++ audio callbacks bypass Java JVM overhead for ultra-low output latency.
- **AEC Feedback Protection:** Acoustic Echo Cancellation suppresses speaker output from feeding back into microphone in full-duplex.

COMMUNICATION & ALERT MODES

- **Push-to-Talk (PTT):** Walkie-talkie mode; press to buffer audio, VAD finalizes sentence on pause, instant burst packet transport.
- **Continuous Phone Mode:** Hands-free conversational streaming with active bi-directional AEC suppression on both devices.
- **Distress Non-Stop Alert:** High-priority alert payload triggers highest permitted / configured alert volume via Android Audio Focus.
- **Architecture Model Matrix:** Whisper (High accuracy / heavy), Vosk (Lightweight), IndicConformer (Streaming + Indic-optimized - Selected).

TECHNOLOGY STACK ARCHITECTURE

[Mobile OS] Kotlin, Android NDK (C++20), AAudio | **[Speech AI]** IndicConformer, Silero VAD, VITS | **[Edge Inference]** Sherpa-ONNX, ONNX Runtime INT8 | **[Audio Engine]** AAudio Native, Oboe, WebRTC AEC | **[P2P Network]** Wi-Fi Direct (P2P), BLE, Cap'n Proto |

FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

➢ **Offline Edge Inference:** Sherpa-ONNX C++ runtime enables low-latency execution of IndicConformer & VITS on arm64-v8a Android devices without cloud APIs.

➢ **Streaming Low Latency:** Chunk-based STT streaming with acoustic sentence boundary detection achieves near real-time text emission (<1.5s total target).

➢ **Thermal & Power Management:** Silero VAD gating prevents continuous neural inference, maintaining low idle CPU usage (<2%) and battery preservation.

RESOURCE & MODEL FEASIBILITY

➢ **AI4Bharat Indic Ecosystem:** Leverages open-source IndicASR & IndicTTS architectures pre-trained on 10,000+ hours of Indic speech data.

➢ **Modular Dynamic Language Packs:** On-demand model loading avoids APK bloat; base app remains <25 MB, loading desired language weights dynamically.

➢ **INT8 Model Optimization:** Dynamic INT8 quantization compresses acoustic models by ~75% (research reference: ~180MB ASR / ~25MB TTS footprint).

VIABILITY & OPERATIONAL READINESS

➢ **Disaster & Defense Resilience:** Ensures life-saving voice links during cyclone/flood telecom blackouts and remote tactical field operations.

➢ **Commodity Hardware Cost:** Functions entirely on standard commercial low/mid-range Android smartphones without requiring expensive satellite radios.

➢ **Modular Mesh Bridge Extension:** Extensible architecture capable of bridging over BLE-connected LoRa radio nodes for multi-kilometer coverage.

TECHNICAL CHALLENGES & MITIGATION

➢ **Acoustic Noise / Distortion:** Mitigation: Native RNNoise / WebRTC pre-processing + fine-tuned noise-robust IndicConformer checkpoint.

➢ **TTS Audio Feedback Loop:** Mitigation: Native Acoustic Echo Cancellation (AEC) and hardware audio routing suppress mic self-capture.

➢ **Packet Loss on Low-Bitrate:** Mitigation: Lightweight binary sequence headers + checksum validation + fast localized retransmission.

36-HOUR REALISTIC IMPLEMENTATION ROADMAP

PHASE 1 (0–12 Hrs): Core Transceiver Loop
2-Device P2P link (Wi-Fi Direct/BLE), AAudio capture, baseline Indic STT + TTS pipeline, Push-to-Talk verified.

PHASE 2 (12–24 Hrs): Edge Optimization
Silero VAD integration, INT8 quantization benchmarks, Latency profiling (RTF <0.25 target), AEC loop tuning.

PHASE 3 (24–36 Hrs): Scaling & Alert Mode
Multi-language expansion (10 Indic scripts), Distress non-interruptible alert broadcast, field test demo.

IMPACTS & BENEFITS

STRATEGIC IMPACT & EVALUATION TARGETS

- **Ultra-Low Bitrate Efficiency (20% Eval):** Replaces 64 kbps continuous audio stream with <150-byte semantic text packets. (Illustrative advantage: >98% transmission footprint reduction).
- **Speech Accuracy & Intelligibility (40% Eval):** State-of-the-art Conformer ASR for low Word Error Rate (WER) across noisy conditions + natural phonetic synthesis via VITS TTS.
- **Low Latency Bounds (20% Eval):** Speech-to-STT completion <600ms, TTS Real-Time Factor ~0.15–0.25 on ARMv8, Speech-to-Speech total delay ~1.5s (prototype benchmark target).
- **10 Indian Languages & Disaster Inclusion:** Hindi, Marathi, Gujarati, Tamil, Telugu, Kannada, Malayalam, Bengali, Odia & English support for non-literate citizen accessibility.

FEATURE COMPARISON: CONVENTIONAL VS ITANTRA

Evaluation Metric / Capability	Conventional Voice	iTantra Transceiver
Transmission Payload	Raw Audio Stream (>64 kbps)	**Semantic Text (<1 kbps) ✓**
Zero Internet / Offline	Limited / Cloud ASR Dependent	**100% Fully Offline Edge ✓**
10 Indian Languages	Inconsistent / Fragmented	**Native Indic Pipeline ✓**
Hardware Compatibility	Requires High Bandwidth	**Low/Mid-Range Phones ✓**
Push-to-Talk (PTT)	Analog / Continuous Stream	**Burst Semantic Packet ✓**
Full-Duplex + AEC	Acoustic Feedback Issues	**Native C++ AEC Engine ✓**
Non-Interrupt Alert	Standard Notifications	**Max-Volume Audio Focus ✓**

HOW ITANTRA WORKS: 3 OPERATIONAL WORKFLOWS

MODE 1: WALKIE-TALKIE (PTT)
Press PTT → Stream Speech → VAD Sentence Formation → Transmit Packet → Receiver TTS Synthesis → Voice Playback.

MODE 2: CONTINUOUS PHONE MODE
Hands-Free Auto-VAD Detection → Real-Time Streaming STT ↔ Bi-directional Text Packets ↔ VITS TTS + AEC Filter.

MODE 3: HIGH-PRIORITY DISTRESS ALERT
Distress Text Ingestion → Instant Priority Queue → Bypasses DND / Max Configured Alert Volume Announcement.

RESEARCH AND REFERENCES

RESEARCH & TECHNICAL FOUNDATION

- **Acoustic-Semantic-Acoustic Transceiver:** Replaces bandwidth-heavy voice streaming with semantic token serialization over low-bitrate physical links.
- **Streaming Conformer RNN-T/CTC:** Enables instant token emission within ~200ms chunks on mobile ARM CPUs without full-sequence buffer latency.
- **Lightweight VAD Speech Gating:** Silero VAD integration stops idle inference, eliminating thermal throttling and background noise transcription.
- **Indic Transliteration & Phonemes:** IndicXlit script mapping enables cross-dialect phonetic accuracy across 10 major Indian language families.

WHY ITANTRA STANDS OUT

- **Semantic vs Waveform Transmission:** Transmits language semantics rather than audio waveforms, achieving ultra-resilient communications on congested channels.
- **Zero-Cloud Edge Multilingualism:** Designed natively for 10 Indian languages with local transliteration and on-device synthesis.
- **Unified Transceiver Architecture:** Single unified mobile codebase handles PTT walkie-talkie, full-duplex phone, and emergency alert broadcasts seamlessly.
- **Indigenous Space & Defense Alignment:** Directly addresses ISRO disaster management and tactical off-grid communication challenges.

PROJECT REFERENCES & AUTHORITATIVE CITATIONS

1. **ISRO Disaster Management Support:** Disaster communication requirements and satellite payload standards. [https://www.isro.gov.in/DisasterManagementSupport.html]
2. **AI4Bharat IndicConformer & IndicASR:** State-of-the-art multilingual speech recognition for Indian languages. [https://ai4bharat.iitm.ac.in/areas/asr/]
3. **AI4Bharat IndicTTS & IndicXlit:** Lightweight Indian language text-to-speech & transliteration engines. [https://github.com/AI4Bharat/IndicTTS]
4. **Sherpa-ONNX (k2-fsa):** High-performance C++ edge inference runtime for next-gen ASR/TTS. [https://github.com/k2-fsa/sherpa-onnx]
5. **VITS & Piper Neural TTS:** Lightweight end-to-end conditional variational speech synthesis on ARM. [https://github.com/rhasspy/piper]
6. **Android NDK AAudio Architecture:** High-performance low-latency audio callbacks and routing on Android. [https://developer.android.com/ndk/guides/audio/aaudio]
7. **ONNX Runtime Mobile Optimizations:** INT8 dynamic quantization and memory-mapped model execution. [https://onnxruntime.ai/docs/performance/model-optimizations/]

FUTURE EXTENSION: NAVIC INTEGRATION

SATELLITE DISTRESS BROADCAST

➢ **NavIC Messaging Service:** Direct integration with indigenous NavIC short-message satellite broadcast receivers for national early warning and tsunami alerts.

➢ **Local Language Re-synthesis:** NavIC satellite text alerts received by iTantra node are instantly synthesized into local regional language voice alerts.

➢ **Illustrative Satellite Bridge:** NavIC Satellite Alert → BLE Dongle → iTantra Edge App → Offline Indic TTS → Non-Interrupt Loudspeaker Announcement (Future Roadmap).